

18/11/2014 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
 Semester: July 2014 - November 2014

Max. Marks: 100**Duration: 3 hrs****End Semester Exam**

Class: SE

Semester: III

Name of the Course: Data Structures

Branch: Computer

Course Code: UCEC303

Instructions:

- (1) All Questions are Compulsory
- (2) Make Suitable assumption whenever necessary and justify the same.
- (3) Figures at the right indicate full marks.
- (4) Specify the question numbers correctly.

Question No.		Max. Marks
Q1	a) Explain Linear and Non-Linear data structures with example. Compare the data structures used to implement the ADT list. b) Show the formation of AVL tree for the following elements 21, 34, 5, 70, 18, 55, 65, 75, 85, 46, 36, 26, 16, 6 Or b) Write a program to accept a graph and perform breadth first traversal of the graph.	(10 + 10)
Q2	a) Write a program to create a sorted doubly list and perform following operations on it. i) Insert into the list ii) Delete from the list iii) Search in the list Or a) Write a program to delete a node from BST considering all the cases b) Write a program to implement input restricted deque. Or b) Write a program to implement two stacks using array.	(10 + 10)
Q3	a) Write the program to convert infix expression to postfix form. b) Using Huffman's algorithm finds the code of symbols whose frequencies are given below.	(10 + 10)

Symbol	Frequency	Symbol	Frequency
E	15	U	5
T	12	I	4
A	10	D	4
O	08	M	3
R	07	C	3
N	06	Q	2
S	05	K	2

Or

b) Explain Johnson's algorithm with an example.

Q4

a) Write the program to perform quick sort. Show the steps with the example

(10 + 10 + 10)

Or

a) Write a program to implement index sequential search. Explain with an example

b) Show the hash table entries (size=19) for the given data using linear probing, quadratic probing method

224562, 137456, 214562, 140145, 214576, 162145, 144467, 199645, 234534, 144728, 194126

c) Sort the following elements in ascending order using radix sort
25, 57, 48, 37, 12, 92, 86, 33

Q5

(10)

a) Identify the appropriate data structure to check whether a string is palindrome or not. Justify the same. Write the algorithm for this operation

Or

b) Identify the appropriate data structure, given the String containing arithmetic expression, to check whether opening and closing parenthesis are well formed. Justify the same. Write the algorithm for this operation